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Circular Design for Value-added Remanufactured End-of-Life composite material via additive manufacturing technology

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Abstract

In a context in which sustainability is going to assume an ever greater role in manufacturing and product development, recycling and reusing end-of-life materials appear as some of the major challenges in technology fields. Currently, a fair amount of methods and processes has been already developed, in order to give a second life to most of conventional materials, such as metals or thermoplastic polymers. Despite the increasing in glass fibre reinforced polymer composites structural applications (i.e. automotive, energy, constructions), it seems that their recycling and reusing are almost unexplored fields, due to the intrinsic multi-material nature. In fact, opposite to thermoplastic polymers, glass fibre reinforced composite materials are commonly composed of a thermoset matrix which cannot be melted and re-shaped after curing.

How could composites be recycled, in a circular economy model? Is there a way for their remanufacturing, so as to obtain value-added and long-lasting products? Situated within “FiberEUUse” European Project, the research’s main aim is to extend the life cycle of glass fibre reinforced polymer composites by using 3D Printing technology. Thus, it allows to produce environmentally sustainable objects with good mechanical properties and complex shapes, potentially suitable for a large range of application fields.

At this purpose, a low-cost Liquid Deposition Modelling technology was used, allowing glass fibre mechanical recycling for Additive Re-manufacturing. More in details, end-of-life shredded composites were adopted as reinforcement of a photo- and thermo-curable material with an acrylate-base resin as matrix: the output can be properly 3d printed by extruding the liquid material before its in situ polymerisation.

Primarily, the experimental work focused on two main aspects: the material characterization, with a remanufacturing process optimisation approach, and its application field analysis, for designing new value-added products. Regarding the first case, it was possible to better understand the way for obtaining the appropriate material composition and process settings, in order to achieve a satisfactory 3d printed outcome, as well as to unveil its mechanical and physical properties. Considering the second one, applications that could take advantages both from composites materials and additive manufacturing technology were investigated.

Through samples design and, consequently, 3d printing realisation, the whole process has been validated: new way to reuse an end-of-life composite material was given by using Liquid Deposition Modelling, promoting the implementation of a circular economy. After process parameters setting, it was possible to perform self-standing 3d prints by modulating the polymerisation directly hardening the reinforced resin after the extrusion, overcoming the current concept of supports in additive manufacturing processes.

Taking into account both the application field and the above-mentioned results, urban design could benefit

from end-of-life composites 3d printing, giving new senso-aesthetic possibilities to designer figures. Considering both material properties and remanufacturing process, street furniture and amusement park structures will be realised directly in situ by a targeted and customised project, as soon as a future process scale-up can be managed.

Keywords: Recycling, Composite Materials, Circular Economy, Additive Re-Manufacturing, 3D Printing

1. Introduction

Currently, sustainability's importance has been strongly increased in last years, especially considering the extensive exploitation of fossil derivatives for human activity and, particularly, for the production of polymer materials (Wackernagel and Rees, 1996; Sikdar, 2003).

As a matter of fact, sustainability assumed a huge role also in manufacturing, considering not only products development and production, but also the disposal at their end-of-life. In fact, a fair amount of methods for end-of-life material recycle has been already developed, considering the most commonly used materials. On the contrary, the recyclability of fibre reinforced polymers seems dramatically poor in comparison with thermoplastics, despite their increasing spreading. In fact, the main issue is linked to their inherent multi-material nature: for this reason, composite wastes have not got experienced recycling processes, yet (Yang et al., 2012).

As regards composite materials, more than one million of tons of composite waste were produced in 2015 within the European Union. Moreover, a huge amount of composite products is reaching their end-of-life, usually from 10 to 30 years depending on their product category, i.e. recreational boats and wind turbine blades (AVK, 2017).

Accordingly, composites recover, recycle and reuse need to be further investigated, moving toward a circular economy model (Pickering, 2006; Yang et al., 2012). In order to enhance this change, material should be recovered by the reuse, recycling, repairing and remanufacturing for reducing waste and avoiding materials downcycling (Oliveux et al., 2015). At this purpose, new composites recycling methods have been developed paying close attention to high-quality recovering, although intrinsic material problems: mainly, scale-up problems are linked to equipment costs and energy demands, outweighing the economic benefit. In a circular economy approach, these issues should be overcome (Mativenga et al., 2017).

Inevitably, the scale-up trouble is also referred to glass fibre reinforced composites (GFRCs) due to their increasing fields of application. Investments can indeed overcome the intrinsic value of recovered products, avoiding real recycling technologies adoption, in contrast to carbon fibres recycle (Wang et al., 2018).

Although the Europe GFRCs production volume is at least one order of magnitude higher than carbon fibre reinforced polymers (CFRPs), GFRCs recycling and reusing seems almost unexplored, especially

comparing thermoset matrix behaviour to thermoplastics (AVK, 2017). Actually, their recycle is possible adopting mechanical grinding and pyrolysis, while the reuse for new products can be made by splitting, hot and cold crushing and hot forming (Åkesson et al., 2012; Adams et al., 2015). In industrial application field, GFRCs recycle can be achieved by mechanical recycling, obtaining different qualities and dimension of recyclates, in order to use them in different applications and with several production techniques (Beauson et al., 2014; Shuaib & Mativenga, 2016).

With regard to other technologies, additive manufacturing (or 3D printing) has been emerging as a good solution for composite material use, manufacturing new products through a layer-by-layer material addition (Wang et al., 2017). Thanks to its several advantages (i.e. shape complexity and freedom, customization, cost reduction in product development and scraps reducing), 3D printing seems to fit well with composites recycling, in order to produce filament or 3D printable inks for low cost 3D printers (Ostuzzi et al., 2015; Griffini et al., 2016; Cruz Sanchez et al., 2017; Mohan et al., 2017; Tian et al., 2017; Mantelli et al., 2019).

Similarly, additive manufacturing could enhance circular economy model diffusion, generating new value in 3D printed products (Despeisse et al., 2017). At the same time, design methods allow to put these models into practice (Moreno et al., 2016) and new related application fields have been developing in last years (Rayna and Striukova, 2016; Ngo et al., 2018).

For all the above reasons, a new way for GFRCs re-manufacturing via 3D printing for new design applications should be the missing gap in circular design approach real application: in this regard, the present research's principal aim is to link these elements in a single process.

In this work, an additive re-manufacturing process for GFRCs recycle was presented and redefined, in order to reuse shredded solid recyclates GFRCs. Through an UV-assisted 3D printing process, a 3D printable material (or ink) made by an acrylic-based photo- and thermo-curable resin and wind blade turbines glass fibres thermoset composites was first developed. Then, the material was optimized in terms of different component percentages, printability of the extruded inks, UV conversion of printed cured materials, rheological properties and mechanical behaviour.

Later, surface finishing and senso-aesthetic properties were examined and, at the end, new environmentally sustainable objects for urban design applications were presented, in order to make them real thanks to a future process scale-up.

2. Methods

3D Printable materials were mainly composed by a photo- and thermo-curable acrylic-base resin matrix and recycled GFRCs powder reinforcement (or recycle GFRCs), mixed in different percentages.

In its turn, the acrylic-base matrix consisted of the ethoxylate bisphenol A diacrylate resin (hereinafter named SE349, purchased from Arkema and locally distributed by Came S.r.l., Italy). Other components in different percentages were added to the matrix as polymerization and filler addition enhancements:

Butanediol dimethacrylate as filler percentage enhancer (BDDMA, purchased from Sigma-Aldrich, Italy), Dicumyl peroxide as thermal activator (purchased from Sigma-Aldrich, Italy) and Ethyl phenyl(2,4,6-trimethylbenzoyl) phosphinate as photo activator (hereinafter named TPO-L, purchased from Lambson Limited, UK).

SR349 and BDDMA percentages were investigated during the experimentation, while TPO-L and Dicumyl peroxide were added using fixed proportions (3% wt. of TPO-L and 0,3% wt. of dicumyl peroxide) investigated by previous works (Mantelli et al., 2019).

Recycled GFRCs derived from Siemens Gamesa Renewable Energy S.A. end-of-life wind turbines. In order to be suitable for additive remanufacturing, turbine's scrap GFRCs had to be shredded up to become fine powder. For the following experimentation, Recycled GFRCs powder was shredded and supplied from Centro Nazionale di Ricerca (CNR), Italy.

By thermogravimetric analysis (TGA) and, consequently, its curves evaluation, it was possible to define the glass content of the recycled powder, here shown in Table 1. Each raw material was used as received.

Table 1. Glass Fibre content (estimated by thermogravimetric analysis), mean fibre length(*l*), mean fibre diameter (*d*) and geometry factor (ζ) for recycled GFRC powder (estimated by scanning electron microscopy analysis).

	Glass Fibre Content (% wt.)	Fibre length, <i>l</i> (μm)	Fibre diameter, <i>d</i> (μm)	Geometry factor, ζ (2* <i>l</i> / <i>d</i>)
Recycled GFRC powder	70 ± 1	34.4 ± 47.7	13.5 ± 6.0	4.6

Formulations were obtained by mixing the matrix components in the established percentages with a magnetic stirrer at room temperature for 2 hours. Afterwards, Recycled GFRCs powders were incorporated to the resin system: initially, they were manually inserted and mixed (for less than a minute), in order to include the volatile powders into the matrix. For obtaining a homogeneous and printable material with a good repeatability, the material needs to be successively mixed by using a mechanical stirrer equipped with a shear stress impeller for 2h at 250 rpm or with a Brabender mixer (C.W. Brabender Instruments, Inc, US) equipped with a roller blade for 45 minutes at 40 rpm. Through this process, it was possible to add a major percentage of recycled GFRCs powder into the acrylic based matrix as presented by previous works (Mantelli et al., 2019). Figure 1 shows the formulations obtained and here studied either with the Brabender mixer or the mechanical stirrer, made by increasing weight concentrations of recycled GFRCs powders and varying BDDMA percentages: for short, the 3D printable material formulations will be referred to as XDYR (where X is the BDDMA reactive diluent concentration and Y the recycled GFRCs percentage).

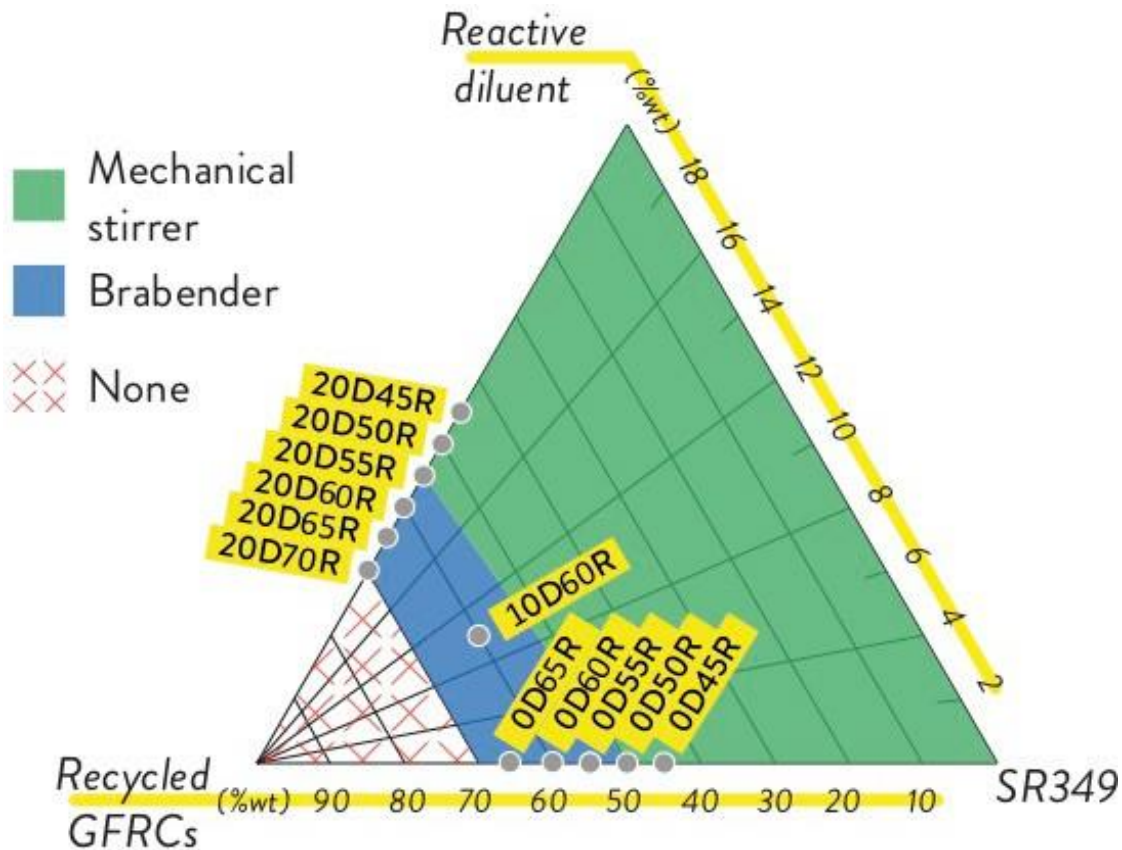


Figure 1. Composition graph: the analysed 3D printable ink formulations are highlighted in yellow. The green and blue areas respectively indicate the compositions mixed with a mechanical stirrer or a Brabender mixer. Finally, the red crosses area indicates material formulations with inadequate processability.

For 3D printing tests, a 3Drag FDM 3D printer (supplied by Futura Group Srl, Italy) was modified as in previous works (Griffini et al., 2016; Mantelli et al., 2019), becoming a Liquid Deposition Modelling (for brevity, LDM) 3D printer, suitable to deposit low viscosity liquids layer-by-layer from a non-heated nozzle linked to a syringe extruder, moved by an endless screw system. The 3d printer was also equipped with three LEDs with a light emission peaked at 395 nm, controlled by a dimmable driver LED DI0001LE, supplied by Tecno Switch, Italy, in order to start material's in situ polymerization.

Considering 3D printer modification, custom pieces were designed by using "Fusion 360" CAD software (Autodesk, US) and "Slic3r PE 1.41.3" open source slicing software (Prusa Research, Czech Republic) and 3D printed with Prusa i3 MK3 FDM 3D Printers (Prusa Research, Czech Republic).

Primarily, LDM 3D printer was used for UV-3D printing samples (process parameters definition and surface finishing test), material and mechanical characterization specimens, and design application scale model. In this case, samples and product concept 3d models were designed using "Solidworks" CAD software (Dassault Systèmes, France) and sliced for LDM process with "Cura" open source slicing software (Ultimaker B. V., Holland).

Regarding 3D printing fixed parameters, print speed ranged between 5 and 10 mm/s and layer height changed between 0,25 and 1 mm, using a stainless steel conic nozzle with a diameter of 1.04 mm.

3D Printable material physical and mechanical characterization was made by combining different kind of test (TGA, rheological tests, photo-DSC, tensile test), giving a feedback both for 3d printability and design applications. Considering the material, each kind of test was performed at a different step.

First, TGA was performed on recycled GFRCs powders with TA INSTRUMENTS Q500 TGA (TA Instruments, Inc, US), heating the samples from 25°C to 800°C with a 25°C/min heating rate under an air environment.

Meanwhile, SEM images were realized with Cambridge Stereoscan 360 (Cambridge Instrument Company Ltd, UK). Moreover, all the SEM images were obtained using a secondary electron probe and sample surfaces for the analysis were previously prepared with a physical vapour deposition of gold for 1 minute. Then, rheological tests were performed on non-polymerized 3d printable material with Kinexus DSR (Malvern Panalytical Ltd., Uk): before the measurements, a constant shear stress was firstly applied for 1 minute (10 Pa for samples with 30% and 45% wt. of recycled GFRCs powders, 1000 Pa for samples with 50, 55, 60 and 65 % wt. of recycled GFRCs powders), followed by 3 minutes' rest period. Moreover, tests were carried out with a 20 mm plate-plate geometry and a 0,7 mm gap and a stress ramp test was performed from 10 Pa to 20000 Pa in 20 minutes, after the rest period (when 1000 Pa pre-shear was applied, the stress ramp measurement started from 100 Pa).

Reactivity measurements were made by Photo-DSC analyses using UV-3d printed samples with a Mettler-Toledo DSC/823e (Mettler Toledo, US), equipped with Lightningcure LC8, Hamamatsu Photonics, Japan. Mainly, tests were performed exposing the samples for 3 minutes to a 365 nm UV radiation (607.7 mW/cm²) twice and the difference in enthalpy between first and second runs was used to measure UV conversion. For performing DSC tests in N₂ environment, heating/cooling ramps from 25°C to 250°C with a 20°/min heating rate were used: so, the exothermic heat flow caused by unreacted groups polymerization was measured in the first ramp, while material glass transition temperature was evaluated in the second one.

In order to perform tensile mechanical properties, UV-3D printed samples had to go through photo and thermal treatment, avoiding partially polymerized areas. Therefore, UV post-curing treatment was performed using a UV chamber Polymer 500W (Helios Italquartz S.r.l., Italy) equipped with a UVA emittance mercury vapour lamp type Zs (950 W/m²) for 15 minutes each side, while thermal post-curing one was carried out by simply heating samples in an oven for 2 hours at 140°C.

Through a Zwick Roell Z010 (ZwickRoell GmbH & Co. KG, Germany) equipped with a 10 kN cell load, tensile mechanical properties were obtained, following the ASTM standard test method D638 – 14 (2014) and D3039/D3039M-17(2017). In the experimentation, specimens had a gauge length of 40 mm, a width of 10 mm, an overall length of 100 mm and a thickness of 2 mm and test speed was 5 mm/min. Moreover, material formulations used for tests was 0D55R and, before tensile tests, specimens were manually polished, in order to eliminate 3D printing typical roughness.

At the end, elastic modulus was predicted by using the Halpin-Tsai model and compared with experimental results.

3. Results and Discussion

3.1. Printability

Through fine process parameters variation, it is possible to 3d print photo- and thermal-curable 3d printable materials (or inks) with different percentage of recycled GFRCs powder and, consequently, various viscosities.

Substantially, printability tests consisted in finding the optimal parameters for 3D printing different material formulations: the main change in each composition is the recycled GFRCs quantity mixed with the matrix, affecting material extrusion's aptitude.

Very first tests were made by 3d printing tensile test 3d model, with a 100% infill and 0,25, 0,5 and 1 mm layer height, starting from a 20D55R formulation. At the beginning, reactive diluent concentration was fixed according to previous works (Mantelli et al., 2019), while recycled GFRCs powder quantity was incremented during mixing step. In fact, material behaviour appeared inappropriate for LDM: during the extrusion trials, the formulation flowed in an uncontrolled way, depositing too much material and causing both lateral spreading and geometry inaccuracies.

Regarding 3d printing quality, efficiency and extrusion ease, some printability issues were also linked to the previous LDM 3D printing system used: at this purpose, a new extruder system has been redesigned, focusing on a new stiffer structure. In addition, syringe (with capability fro 10 ml to maximum 60 ml) is better fixed to the extrusion system, avoiding syringe plunger's torsion during the process and improving flow stability. UV Led support system was not subject to significant variations from the previous version, that already reached a homogeneous distribution of the UV radiation on the 3D printed piece. In Figure 2, the new LDM extruder system scheme is briefly represented.

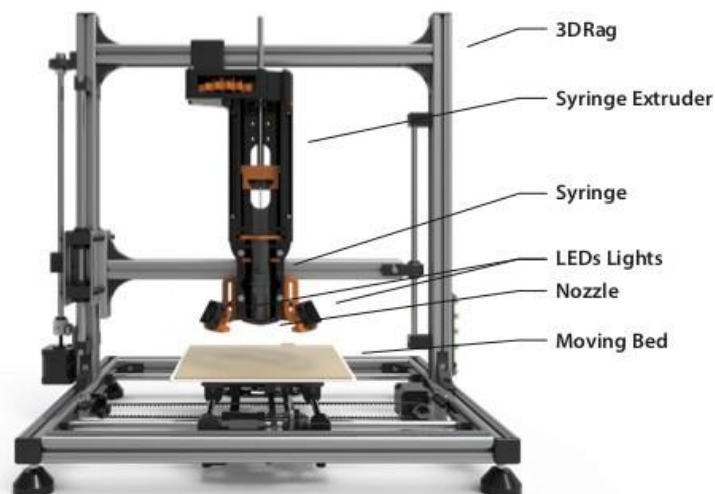
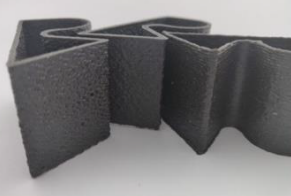


Figure 2. Representation of the UV 3D printing modified in this work.

After the 3D printing upgrade, tests on printability started again with the 20D55R formulation and, in order to set the stepper motor motion on the endless screw (measured in rpm): this operation allows to control the extruded material quantity, matching the GCode extrusion length to the one performed by the machine. In fact, to 1 mm extrusion set in the GCode should correspond 1 mm real extrusion made by the 3D Printing. By increasing the recycled GFRCs powder amount, the material printability was strongly decreasing, even if the percentage was just slightly augmented: theoretical maximum content of recycled GFRCs is 60% vol., that corresponds to 70% wt. (). Furthermore, BDDMA reactive diluent addition was investigated at different percentages (0, 10 and 20% wt.) and, considering the same amount of recycled GFRCs, avoiding its use improved 3D printing quality, despite of previous work. In fact, formulations with reactive diluent trend was to be easily extruded at the beginning, leaving a material with a higher GFRCs concentration in the syringe, as if the matrix was divided from the filler.

According to Table 2, the best 3D printable material was 0D55R formulation, that grants a good flow and less nozzle obstruction if compared to the other compositions.

Table 2. *Printability tests with different material formulations and printing parameters, i.e. flow percentage, speed and layer height: (a) 20D55R tensile test specimen, (b) 0D55R finishing samples, (c) 20D60R tensile test specimen, (d), 0D60R tensile test specimen, (e) 20D65R tensile specimen.*

Sample	20D55R	0D55R	20D60R	0D60R	20D65R
Flow (%)	90	100	120	105	140
Speed (mm/s)	10	9	7	5	5
Layer Height (mm)	0.25 - 1	0.25 – 0.75	0.5	0.5	0.5
Printability	**	***	*	**	*
					
	(a)	(b)	(c)	(d)	(e)

In addition, other process parameters were modified, in order to improve layer adhesion, reduce material voids and nozzle obstruction during 3d printing: LED intensity was from 50% to 70% decreased thanks to the dimmable driver, giving more time to the extruded material for depositing and adapting on the previous layer before its in situ polymerization. Hereafter, this operation will be called “under-polymerization” and its intensity will be modified between according to 3D model geometry and infill/shell parameters.

Due to bottom layer importance for FDM and LDM processes, matching the under-polymerization with a fine z axis calibration contributed to 3D printing quality improvement, achieving good results using spiral slicing option, single and double wall thickness and infill presence.

Contrary to Griffini and coworkers study (Griffini et al., 2016), material photo-polymerization was not affected by recycled GFRCs presence: in fact, all the formulation had a dark grey colour appearance, that could reduce photopolymerization efficiency. However, this unexpected behaviour could be deeply studied in next works.

3.2. Material Characterization

The recycled material was first studied: the glass fibre content, evaluated from TGA measurement, is equal to 70±1%wt. More specifically, the result confirms the high content of glass fibres expected in a composite part and also the conservation of the material constituent during the shredding process. Then, the recycled powder was studied with SEM micrograph: a representative image is visible in Figure 3. The glass fibres were measured from the images and the result will be used later for the young's modulus prediction with Halpin-Tsai model.

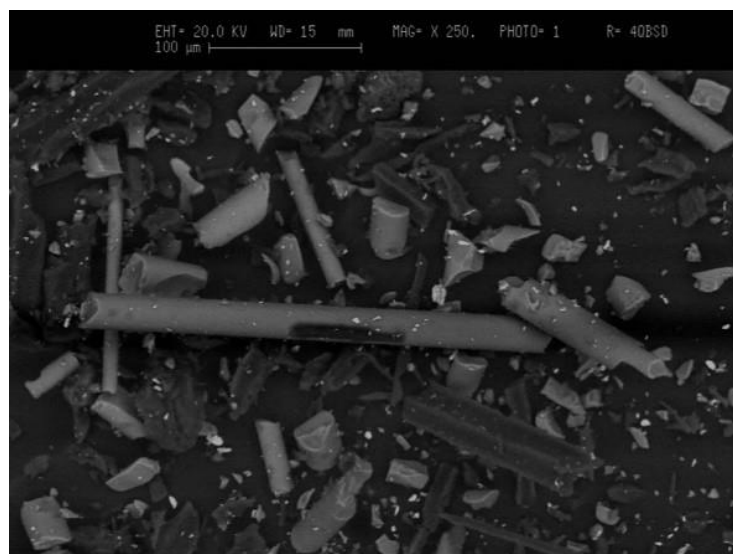


Figure 3. SEM micrograph of the recycled GFRC showing the presence of different glass fibres with different diameters.

The 3D printable ink rheology test visible in Figure 4 shows that the material has a pseudoplastic behaviour. The effect of the increased content of recylate higher the pseudoplasticity behaviour, moreover the introduction of a reactive diluent lowers the overall viscosity. From the graph, it is clear that a 5% wt. increase in recycled GFRCs content produce an increase of the apparent viscosity of one order of magnitude. Furthermore, the addition of 20% reactive diluent to the 55% wt. recylate formulation produce a decrease of one order of magnitude to the apparent viscosity. As a consequence, considering that the best performing 3D printable material has an apparent viscosity of 200 Pa·s, higher viscosity materials produce higher resistance to flow and more difficulties in the extrusion process. On the contrary, lower viscosity inks are difficult to be extruded in a controlled manner. As an open point, the separation of the resin system and

recycled GFRCs powder, particularly for the inks containing the reactive diluent, has to be further studied.

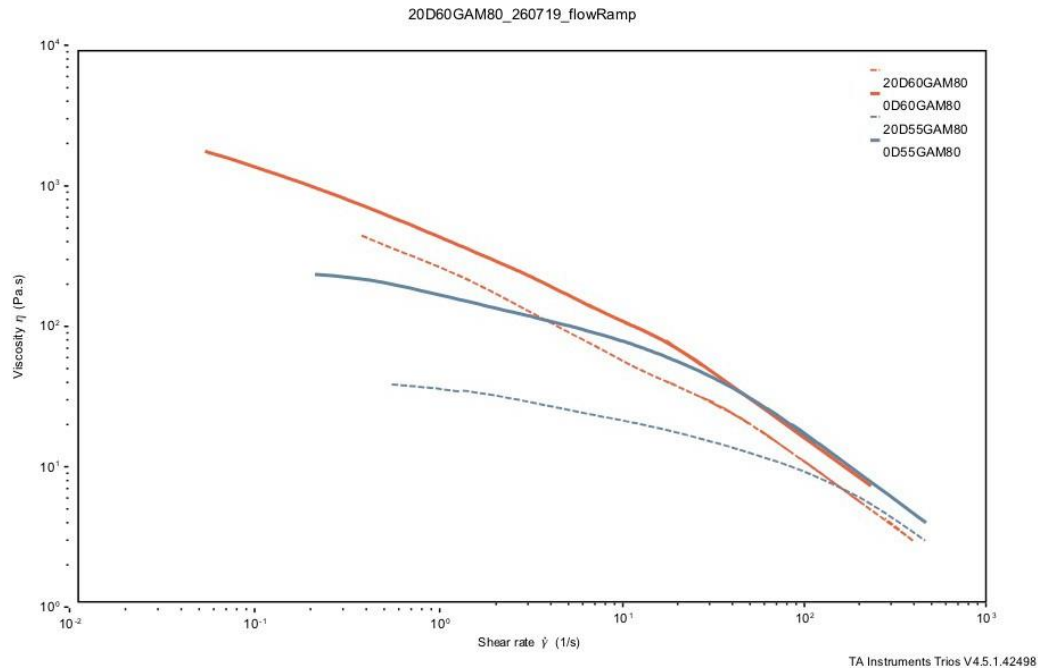


Figure 4. Stress ramp test results for 3D printable materials containing 55 and 60% wt. recyclate both without reactive diluent and in its presence.

Primarily, the UV reactivity of the material was studied in order to understand if the concentration of recycled GFRC has an influence on its reactivity. The UV-conversion related to UV exposure was determined by measuring the exothermic enthalpy produced during UV-crosslinking for the neat resin and for the 3D printable ink as follows:

$$UV_{conv.} = \frac{\Delta H_{sample}}{\Delta H_{neat resin}} \times 100 \quad (1)$$

As can be seen in Figure 5a, the 3D printable material has a quite high UV-conversion even at higher concentrations. As a consequence of the high UV-reactivity during the 3D printing process, the 3D printable material can easily clog the nozzle. To overcome this problem, the UV-LED power was reduced by at least 50%. Controlling the crosslinking kinetics of the 3D printable material flowing out of the nozzle, it was possible to obtain a good shape retention without clogging the nozzle. As discussed in the previous chapter, the UV power control has a consequence also on printability and 3D printed sample quality.

After 3D printing, the UV-DSC test was repeated and the residual conversion (residual enthalpy of 3D printed samples) of a 3D printed sample visible in Figure 5a is very low. Therefore, this result confirms that even in under-polymerization conditions the UV-conversion during the 3D printing process is higher than the 60%, which is a sufficient value for shape retention.

In order to complete the UV-conversion, a UV-post curing cycle is applied to the 3D printed sample.

According to Figure 5b, an exothermic enthalpy is clearly visible after the UV-post curing, which is no more visible after the thermal post curing treatment. Thus, it is confirmed the thermal post curing cycle need, defined in the previous work (Mantelli et al., 2019). After the whole post curing treatment, it is possible to see that the material has a relatively high glass transition temperature at 100°C.

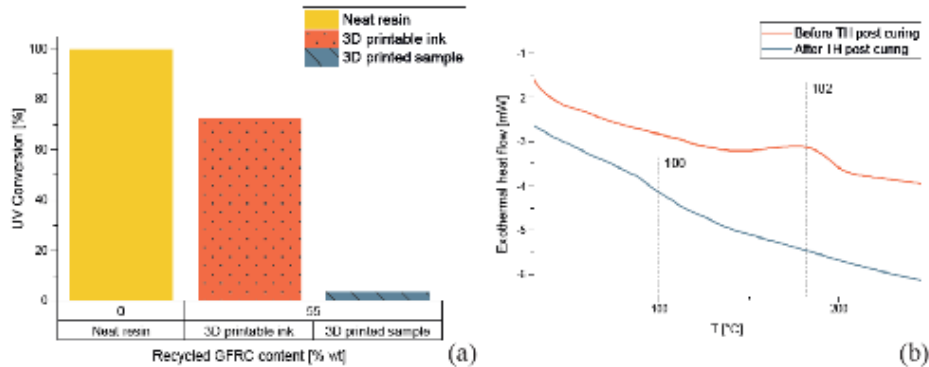


Figure 5. DSC analyses for 3D printable material before and after 3D printing (with different recycled GFRCs percentages): (a) UV conversion of 3D printable materials evaluated by PHOTO-DSC, (b) DSC first heating ramp showing the exothermic peaks' presence for 3D printed samples after a UV post-curing treatment and third heating ramp showing where the glass transition temperature is present.

Taking into account the previous DSC analysis results, the workflow that starts from the 3D printable material mixing to the final object is summarized as follows in the Figure 6. Even though the 3D printed object has already a fixed and manipulable shape due to the UV-assisted process, the UV post curing and the thermal one have a big role in order to complete the material crosslinking.

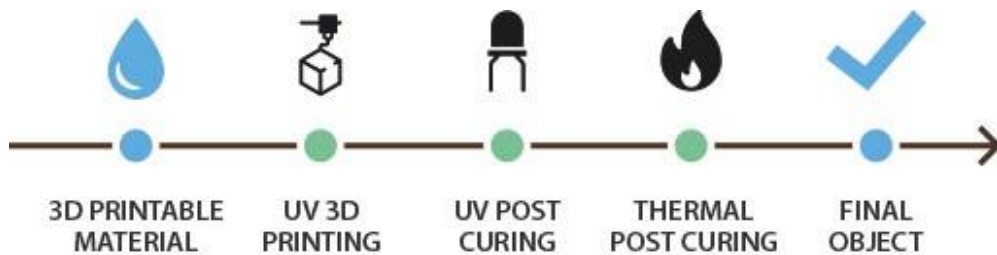


Figure 6. UV 3D printing main steps: 3D printable material mixing, UV 3D printing (obtaining the 3D printed object), UV post curing (UVA 950 W/m² for 30 min), Thermal post curing (140°C for 2 h, obtaining the final product).

Then, the mechanical behaviour of the 3D printed material was studied accordingly to the standards for composites material. Particularly, the results are compared to the neat resin mechanical properties and with the predicted young's modulus calculated with Halpin-Tsai model for aligned fibres (Halpin and Kardos, 1976).

As can be seen from Figure 7, the measured elastic modulus is comparable with the predicted one, confirming the fibres alignment during deposition. Moreover, the two-fold increase in elastic modulus compared to the neat resin is a good result, especially considering the relatively low amount of glass fibres present in the formulation. For the mechanical properties at failure, the results are for a practical point of view similar to the neat resin one.

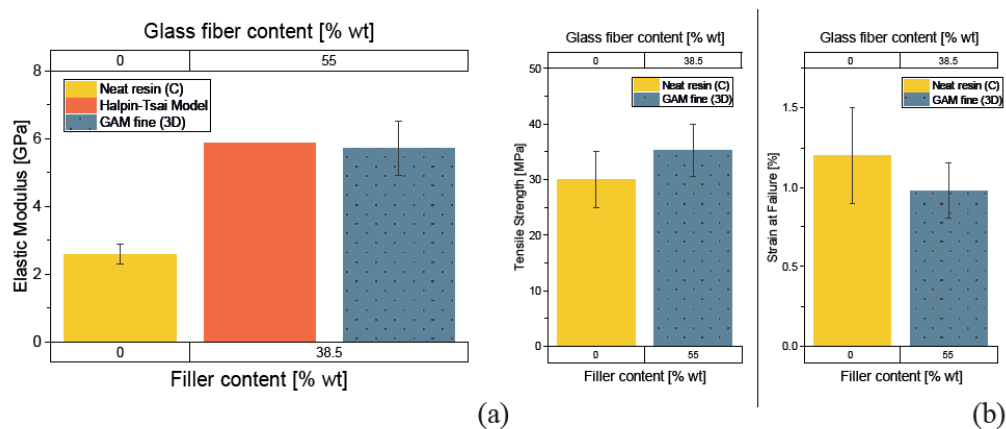


Figure 7. Tensile test results: (a) experimental value of elastic modulus with theoretical values (Halpin-Tsai model), (b) experimental values of ultimate tensile strength, and (c) experimental values of strain at failure.

3.3. Surface Finishing and Senso-aesthetic Features

According to printability tests, surface finishing related to 3d printing re-manufacturing parameters and senso- aesthetic features were investigated.

In literature, material surface finishing importance is well-known, not only for material science, but also in design fields: senso-aesthetic (sensual, aesthetic, and psychological) properties are strictly related to human perception (Hartcher-O'Brien et al., 2019) and, in turn, they are strongly affected by physical and chemical material properties. Furthermore, both vision and touch have a big role in material texture perception (Whitaker et al., 2008).

In addition, manufacturing methods have a big role in surface definition, both considering limits (i.e. roughness, tolerances) and possibilities (i.e. texture definition).

Despite senso-aesthetic properties are subjective and perception could change from an individual to another, previous works have already defined some guidelines (Zuo et al., 2001; Zuo and Jones, 2005). Moreover, material libraries have been assuming an ever great role in design process and product definition, giving physically access to materials samples with different properties and surface finishing (Miodownik, 2007). After defining the best 3D printable material, senso-aesthetic properties were explored together with process parameters by means of 3D printed samples. In particular, 3D model was expressly designed for LDM process features, considering a medium shape complexity, low building times and process parameters

variation ease (overall dimensions: 25x50x30 mm).

Therefore, it was possible to obtain a little surface finishing selection of the 3D printable material made by nine 3D printed and finished samples. As shown in Figure 8, specimens were organized in three main groups: layer height variation (0,25, 0,5, 0,75 mm), printing mode (spiral, two shells, two shells with infill) and post processing finishing (after LDM process, polishing and gel coating, polishing with painting and gel coating). For samples realization, 0D55R material formulation was adopted.

In general, LED settings were adopted accordingly to Printability paragraph and reduced to 70% by the dimmable driver: this permitted to the 3D printable material a better deposition during the extrusion, increasing surface quality. Moreover, decreasing print speed (according to layer height or printing mode) improved 3d printing overall quality, avoiding failures.



Figure 8. Surface Finishing Selection of 3D printable material (0R55D) divided in (a) Layer Height Variation,

(b) Printing Mode and (c) Post Processing Finishing.

Considering layer height variation (Figure 8a), 3D printing layer-by-layer surface finishing is more noticeable in 0,75 mm sample: in the other ones, layer-by-layer effect is strongly reduced by the flow deposition.

Despite LDM layers are thicker than FDM ones thanks to greater nozzle sizes, 0,25 and 0,5 mm samples

have not a defined layer-by-layer surface: on the contrary, layers seemed merged by a surface texture created during the extrusion. More in detail, a random pattern was generated in 3D printing phase. The pattern changed even using the same GCode file and printing settings, too.

In figure 8b, it is shown printing mode change through three 3D printed samples. Even though there is not a significant variation in surface quality, the two shells specimen seemed to have the best layer adhesion, without visible voids. Infill creation was possible, even if it required an accurate travel speed setting for flow deposition improvement.

Finally, post processing treatments on two samples were performed after photo and thermal treatment (shown in figure 8c) and compared with a non-treated one. More in detail, the two 3D printed pieces were manually polished and finished with a transparent gel coating and a white acrylic-based paint with transparent gel coating, respectively. Due to the surface treatments, overall roughness was strongly reduced, improving environmental resistance and changing material's perception.

To sum up, different layer height and printing mode were successfully investigated, changing surface quality and finishing. Although it is not possible to suddenly stop the material flow for moving the nozzle to another position without extrusion issues, LDM process could generate specially designed highquality products.

Focusing on material perception, the combination between the surface random pattern and material colour give to the 3D printed samples a raw stone-material appearance. Especially, each piece seems unique and hand-crafted, generating an extra added value due to its finishing.

Making a comparison between the actual samples and previous works one shown in Figure 9a (Mantelli et al., 2019), it is clear that layer-by-layer appearance is strongly reduced by the material formulation, which is in turn affected by recycled GFRCs powder provided in this work (Figure 9b).

Advantages due to this specific surface finishing will be discussed afterwards, linking them to design applications.

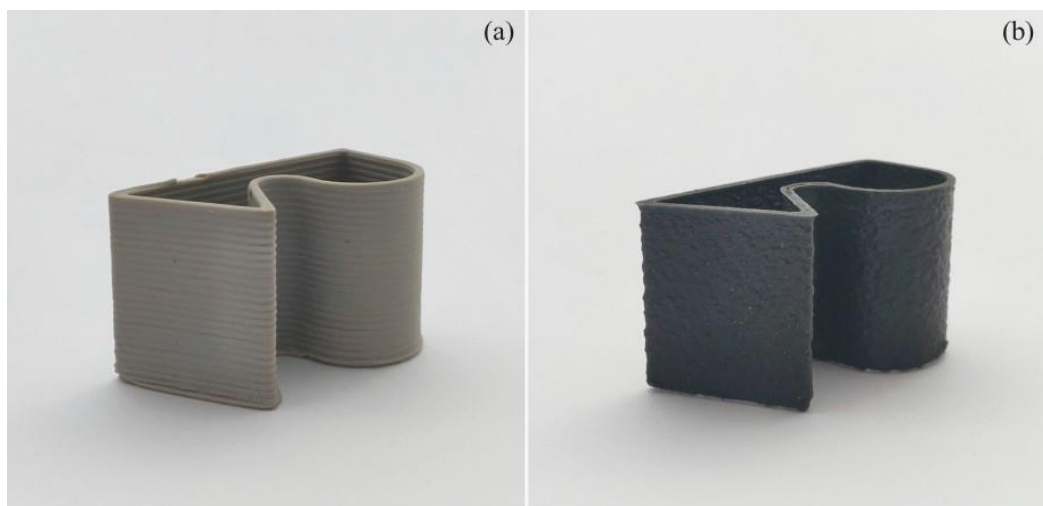


Figure 9. Surface Finish after LDM process of different recycled GFRCs: (a) material formulation from Mantelli et al. (2019), and (b) from this work.

3.4 Additive Re-manufacturing and Design Applications

After surface finishing exploring and senso-aesthetic definition through samples realization, design applications should be investigated, in order to exploit recycled GFRCs powders and 3D printable materials.

Thanks to their properties, new design fields could be reached, enhancing innovation. Therefore, new materials have to be implemented in materials selection, encouraging designers to taking them into account both from a technical engineering and a user-centred design point of view. Moreover, this effort should be done from the early design phases, impacting on life cycle costs (Veelaert et al., 2016).

With respect to a circular economy approach, additive manufacturing materials have been used for reducing products environmental impact by taking advantage of on-demand and in-situ production, reducing transportation costs. At this purpose, several applications could be found in architecture and building field: due to overall product dimensions, manufacturing could be significantly supported. Furthermore, recycled materials have the possibility to generate new value through 3d printing, generating high added-value applications especially from large-scale mass customization (Rayna and Striukova, 2016; Ngo et al., 2018).

Despite full-scale construction via additive manufacturing is still in an emerging phase, construction field and urban design could beneficiate from 3D printing adopting. In fact, material is used in an optimized way, reducing wastes, avoiding damages caused by transportation and, generally, decreasing environmental impact from the initial construction phase to the End-of-life one (Lim et al., 2012; Dixit, 2019).

From a senso-aesthetic point of view, 3D printed samples raw stone-material appearance as investigated above combines well with outdoor construction applications, with similar surface finishing and colours, as well as less layer-by-layer look. As previously mentioned, the random texture and raw stone-material appearance can be easily attributed to buildings and outdoor application fields, reminding other materials that are widely diffused in these contexts, such as ceramic, concrete and stones.

Moreover, this non-3d printed appearance could be advantageous for post processing costs reduction, avoiding expensive treatments in order to improve surface roughness. Actually, outdoor applications will not have a layer- by-layer look, even though any polishing treatment is not performed. In this case, the limit would be due only to 3d printing quality, in its turn obtained by a fine UV and 3D printing parameters setting.

For all these reasons, outdoor architecture, urban design and scenography elements located in amusement parks are suitable for recycled GFRCs printable materials use. In fact, public and private infrastructures made by recycled end-of-life material could generate added-value and, simultaneously, reduce environmental impact. Considering End-of-life composite material annual quantities, construction-scale products seems the best method for reusing a remarkable material amount, avoiding desktop-scale 3d printing pieces.

As a consequence, street furniture, scenography elements and amusement park products made by recycled GFRCs via additive manufacturing will be manufactured directly in-situ, taking advantage both from 3D printing (i.e. customization, complex shapes and low production batches) and composite material properties

(i.e. good mechanical properties, unique senso-aesthetics features). In addition, circular economy key elements could be achieved through the whole process, with significant value addition.

As a proof-of-concept, a scale model of an amusement park main entrance was 3D printed with the 0D55R material formulation, previously designed in other works (Paracchini, 2018): as shown in Figure 10a, a GCode from the STL file was obtained and subsequently printed by UV-assisted LDM process. More specifically, the overhanged entrance was a self-standing structure, that has a tilt angle of 30° with respect to the printing surface. In this case, it was possible to 3D print it without any support, thanks to a fine photopolymerization modulation that directly hardened the material after the extrusion (Figure 10b). After UV and thermal post curing, the 1:50 scale model was assembled (Figure 10c): surface finishing was similar to 3D printed samples here as well.

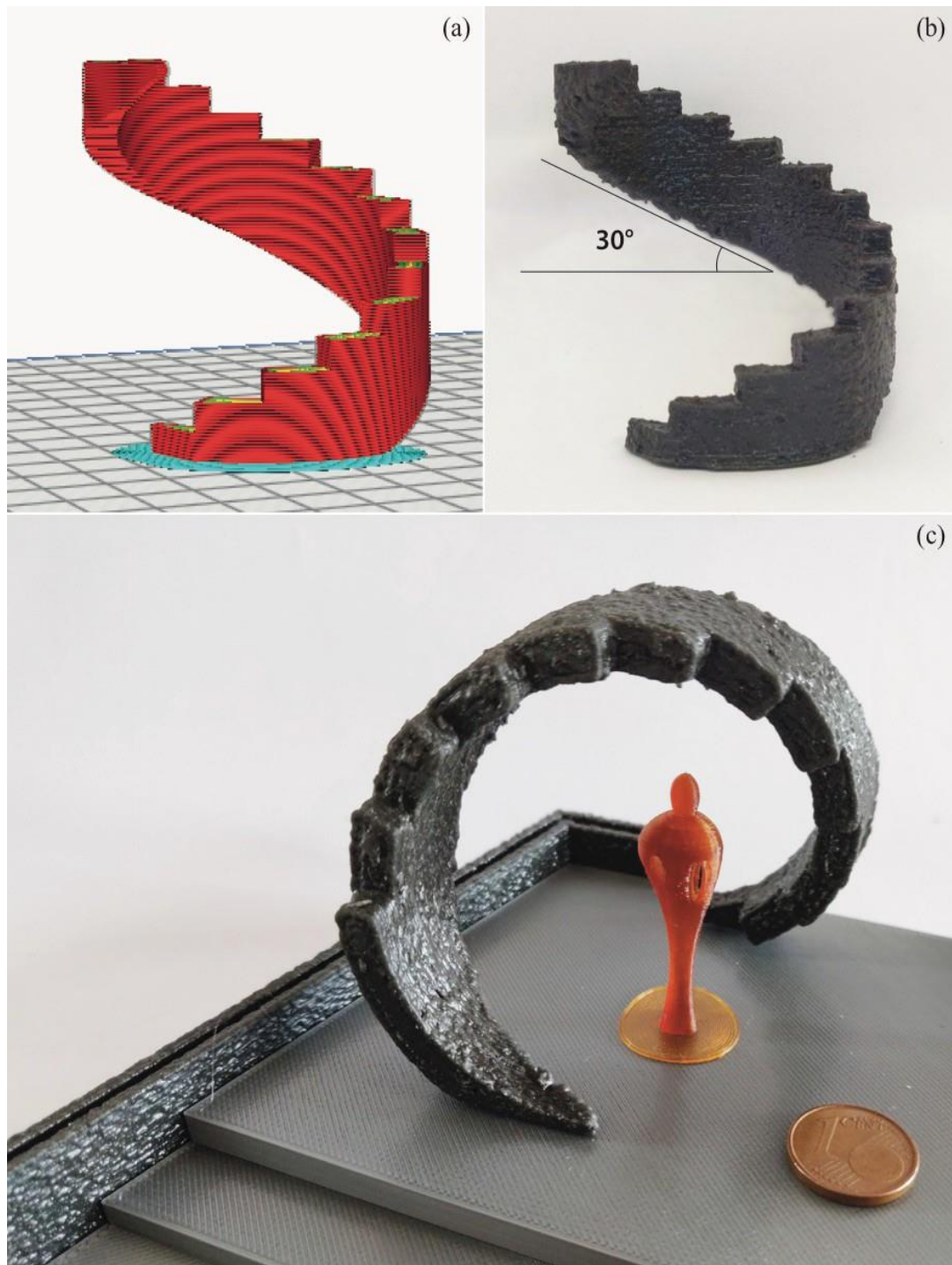


Figure 10. Additive Re-Manufacturing: (a) G-Code preview of the 3D printed structure and (b) the corresponding 3D printed object obtained by the LDM process, (c) showed as 1:50 scale model for amusement park attractions and outdoor architecture application.

As regards process parameters modifications, a new proof-of-concept was performed, in order to validate under- polymerization and led intensity reduction. In other words, layer adhesion improvement was tested by 3d printing a two shell box with a simple shape. After post curing, the box was filled with coloured water and potential water

leaks were checked, in order to verify layer adhesion. In brief, the sample box showed a rudimental water-

tight, that needs to be further analysed in next works.

Accordingly, design fields could be extended to water-based applications, particularly linked with urban design and amusement park structures for decoration purposes, for instance outdoor decorative pools, water park scenography, fountains.

In Figure 11a, different fountain concepts are shown. Despite the material should be tested for allowing human water-contact without health issues, drinkable water fountains could be 3d printed by designing inner hollows for water piping insertion, validating their urban design application.

Then, a 1:20 scale drinking fountain was designed and subsequently 3D printed via UV-assisted LDM technology with the 0D55R material formulation: the overhang complex shape was achieved thanks to the under- polymerization combined with the “on-air” 3D printing mode (Figure 11b). As soon as the LDM scale-up process is tested and validated, further applications and design concept could be developed and widespread.



Figure 11. Additive Re-Manufacturing: (a) Design concepts of water decoration structures for urban design and amusement parks (fountains), and (b) 3D printed 1:20 scale structure obtained by LDM process with 0D55R material formulation.

4 Conclusions

To sum up, a 3D printable material made by a photo- and thermo-curable acrylic-based thermoset matrix and recycled GFRCs powder for UV-assisted 3D printing via Liquid Deposition Modelling (LDM) was developed. As a consequence, an additive re-manufacturing process was redefined, validating UV LED under-polymerization and 3D printing parameter settings. Therefore, a UV-assisted 3D printing of 0D55R material formulation was achieved with a good overall quality.

Then, material characterization of 3D printable inks was performed, defining at first GFRCs powder main features. From 0D55R material formulation, rheological properties were analysed, showing a pseudoplastic behaviour. Moreover, UV and thermal post-curing minimized the residual unreacted groups in 3D printed pieces. Mechanical properties were also investigated, confirming the Halpin-Tsai model prediction.

As regards surface finishing and senso-aesthetic properties, a selection of nine 3D printed samples was realized, considering layer height variation, printing mode and post process finishing: process parameters allowed also to obtain an improved layer adhesion and, consequently, a non-3D printed appearance using specific settings.

In line with recent 3D printing design trends, new design application for urban design and water-contact urban design were found, thanks to scale models and structure realization. Furthermore, self-standing and water-tight structures were obtained by modulating the UV LED intensity during the 3D printing, avoiding supports.

Despite next works are necessary in order to better investigate under-polymerization effects, process scale-up and material characterization, this study leads the way to GFRCs recycling for urban design and construction fields, with unique senso-aesthetic properties for designers.

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